PROPERTIES OF MORPHISMS PRESERVED BY A CARTESIAN FINITE ORDER THICKENING OF THE BASE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-morphisms.tex; no statement or proof step was rewritten, reordered or removed, including the author's own eleventh item 'add more here'.

Editorial note. The two leading context spans are the chapter's own definition of thickenings, first order thickenings, finite order thickenings and morphisms of thickenings (label definition-thickening), and the author's discussion of the filtration of an nth order thickening by first order thickenings that the proof cites at line 379; both define or explain notions appearing in the counted statement, and definition-thickening is therefore not listed among the external references.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-more-morphisms/retained-preamble.tex): a verbatim excerpt of the native Xy-pic setup and of the native theorem declarations and mathematical macros. The native class line, the xr-hyper externaldocument block and the reference/slogan/history display environments are omitted because the assembler supplies a plain amsart wrapper and the excerpt uses none of those environments; no macro was invented or redefined.

Editorial note. Locators of the cross-chapter pointers refer to a verbatim excerpt of the project's own tag list, sources/stacks-more-morphisms/referenced-tags.txt, each line carrying its stable Stacks tag; the same-chapter pointer to lemma-thicken-property-morphisms refers to its own label line inside the pinned chapter file.

Allowed standard prerequisites (editorial).

- Stability of the listed morphism properties under base change. Schemes, Stacks tags 01JY and 02YC, and Morphisms, Stacks tags 01T4, 01TM, 02NK, 01V0, 02GA, 01W4 and 01WL, all displayed at proof lines 358-371. Gives the easy direction of every clause: if f' has the property then so does its base change f along X = S times_{S'} X'.
- Nakayama's lemma. Algebra, Stacks tag 00DV (algebra-lemma-NAK), invoked at proof lines 405, 410 and 430. Turns surjectivity of A to B into surjectivity of A' to B', lifts finite typeness along A'[x_1,...,x_n] to B', and upgrades Omega_{B'/A'} = I Omega_{B'/A'} to Omega_{B'/A'} = 0.
- Reduction of a finite order thickening to first order thickenings. the chapter's own discussion printed as the second context span (source lines 66-85), cited by the proof at line 379. Justifies the induction on the order of the thickening with which the proof may assume that S in S' is a first order thickening.
- Permanence of quasi-compactness, separatedness and universal closedness under thickenings. More on Morphisms, Stacks tag 09ZV (more-morphisms-lemma-thicken-property-morphisms), invoked at proof lines 441 and 447.

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Supplies the remaining ingredients for clauses (7), proper, and (8), finite, once finite typeness and the integral or unramified part is available.
- Fibrewise criteria for quasi-finiteness and for relative dimension d, and the diagonal criterion for differentials. Morphisms, Stacks tags 01TH, 02NJ (morphisms-definition-relative-dimension-d) and 08S2, invoked at proof lines 415-422 and 462-464. Lets clauses (3) and (4) be checked on fibres, and lets clause (9) reduce the monomorphism statement to the vanishing of Omega_{X'/S'}.

Source reference. Original lemma 09ZV: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 270–270. Same-chapter input invoked at proof lines 441 and 447 for clauses (7) and (8); a permanence statement about quasi-compactness, separatedness and universal closedness that the counted proof combines with the finite type clauses it proves itself.

Source reference. Original lemma 00DV: sources/stacks-more-morphisms/referenced-tags.txt, source lines 4–4. Nakayama's lemma invoked three times, at proof lines 405, 410 and 430; a standard algebraic input, while the affine reduction that makes it applicable is part of the counted proof.

Source reference. Original lemma 00RV: sources/stacks-more-morphisms/referenced-tags.txt, source lines 5–5. Identification of Omega_{B/A} with Omega_{B'/A'} modulo I Omega_{B'/A'} at proof line 427; a navigation pointer, the Nakayama step that consumes it being in the counted proof.

Source reference. Original lemma 01T4: sources/stacks-more-morphisms/referenced-tags.txt, source lines 6–6. Base change stability of local finite typeness, displayed at proof lines 364-365 as part of the uniform easy direction.

Source reference. Original lemma 01TH: sources/stacks-more-morphisms/referenced-tags.txt, source lines 7–7. Fibrewise criterion for quasi-finiteness used at proof line 417 to deduce clause (3) from clause (2).

Source reference. Original lemma 01TM: sources/stacks-more-morphisms/referenced-tags.txt, source lines 8–8. Base change stability of local quasi-finiteness, displayed at proof line 366 as part of the uniform easy direction.

Source reference. Original lemma 01V0: sources/stacks-more-morphisms/referenced-tags.txt, source lines 9–9. Base change stability of the vanishing of differentials, displayed at proof line 368; the algebraic translation of clause (5) is in the counted proof.

Source reference. Original lemma 01W4: sources/stacks-more-morphisms/referenced-tags.txt, source lines 10–10. Base change stability of properness, displayed at proof line 370 as part of the uniform easy direction.

Source reference. Original lemma 01WJ: sources/stacks-more-morphisms/referenced-tags.txt, source lines 11–11. Description of finite morphisms as integral and locally of finite type, used at proof line 449 for clause (8).

Source reference. Original lemma 01WL: sources/stacks-more-morphisms/referenced-tags.txt, source lines 12–12. Base change stability of finiteness, displayed at proof line 371 as part of the uniform easy direction.

Source reference. Original lemma 01JY: sources/stacks-more-morphisms/referenced-tags.txt, source lines 14–14. Base change stability of immersions, displayed at proof line 361 as part of the uniform easy direction for clause (10).

Source reference. Original lemma 02GA: sources/stacks-more-morphisms/referenced-tags.txt, source lines 15–15. Base change stability of unramifiedness, displayed at proof line 369 as part of the uniform easy direction.

Source reference. Original lemma 02NK: sources/stacks-more-morphisms/referenced-tags.txt, source lines 16–16. Base change stability of the relative dimension, displayed at proof line 367; clause (4) is then checked on fibres inside the counted proof.

Source reference. Original definition 02NJ: sources/stacks-more-morphisms/referenced-tags.txt, source lines 17–17. Definition of locally of finite type of relative dimension d, cited at proof line 422 for the fibrewise reading of clause (4).

Source reference. Original lemma 02YC: sources/stacks-more-morphisms/referenced-tags.txt, source lines 18–18. Base change stability of monomorphisms, displayed at proof line 362 as part of the uniform easy direction for clause (9).

Source reference. Original lemma 08S2: sources/stacks-more-morphisms/referenced-tags.txt, source lines 20–20. Diagonal criterion for differentials invoked at proof line 462 to finish clause (9) from the vanishing of Omega_{X'/S'}; the surrounding computation of the ideal of the diagonal is in the counted proof.

Definition 0.1. Thickenings.

(1) We say a scheme X' is a *thickening* of a scheme X if X is a closed subscheme of X' and the underlying topological spaces are equal.
(2) We say a scheme X' is a *first order thickening* of a scheme X if X is a closed subscheme of X' and the quasi-coherent sheaf of ideals $\mathcal{I} \subset \mathcal{O}_{X'}$ defining X has square zero.
(3) We say a scheme X' is a *finite order thickening* of a scheme X if X is a closed subscheme of X' and the quasi-coherent sheaf of ideals $\mathcal{I} \subset \mathcal{O}_{X'}$ defining X is nilpotent, i.e., there exists an integer $n \geq 0$ such that $\mathcal{I}^{n+1} = 0$.
(4) We say a scheme X' is an *nth order thickening* of a scheme X if X is a closed subscheme of X and $\mathcal{I}^{n+1} = 0$ where $\mathcal{I} \subset \mathcal{O}_{X'}$ is the quasi-coherent sheaf of ideals defining X.
(5) Given two thickenings $X \subset X'$ and $Y \subset Y'$ a *morphism of thickenings* is a morphism $f' : X' \to Y'$ such that $f'(X) \subset Y$, i.e., such that $f'|_X$ factors through the closed subscheme Y. In this situation we set $f = f'|_X : X \to Y$ and we say that $(f, f') : (X \subset X') \to (Y \subset Y')$ is a morphism of thickenings.
(6) Let S be a scheme. We similarly define *thickenings over S*, and *morphisms of thickenings over S*. This means that the schemes X, X', Y, Y' above are schemes over S, and that the morphisms $X \to X'$, $Y \to Y'$ and $f' : X' \to Y'$ are morphisms over S.

Let $i_X : X \to X'$ be a thickening. Any local section of the kernel $\mathcal{I} = \mathrm{Ker}(i_X^\sharp)$ is locally nilpotent. This gives some control over the structure sheaf of $\mathcal{O}_{X'}$, but technically the class of finite order thickenings $X \subset X'$ is much easier to handle. Namely, if X' is an nth order thinking then we have a filtration

$$0 = \mathcal{I}^{n+1} \subset \mathcal{I}^n \subset \mathcal{I}^{n-1} \subset \ldots \subset \mathcal{I} \subset \mathcal{O}_{X'}$$

and we see that X' is filtered by closed subspaces

$$X = X_1 \subset X_2 \subset \ldots \subset X_n \subset X_{n+1} = X'$$

such that each pair $X_i \subset X_{i+1}$ is a first order thickening over S. Using simple induction arguments many results proved for first order thickenings can be rephrased as results on finite order thickenings.

Lemma 0.2. *Let $(f, f') : (X \subset X') \to (S \subset S')$ be a morphism of thickenings such that $X = S \times_{S'} X'$. If $S \subset S'$ is a finite order thickening, then*

(1) *f is a closed immersion if and only if f' is a closed immersion,*
(2) *f is locally of finite type if and only if f' is locally of finite type,*
(3) *f is locally quasi-finite if and only if f' is locally quasi-finite,*
(4) *f is locally of finite type of relative dimension d if and only if f' is locally of finite type of relative dimension d,*
(5) *$\Omega_{X/S} = 0$ if and only if $\Omega_{X'/S'} = 0$,*
(6) *f is unramified if and only if f' is unramified,*
(7) *f is proper if and only if f' is proper,*
(8) *f is finite if and only if f' is finite,*
(9) *f is a monomorphism if and only if f' is a monomorphism,*
(10) *f is an immersion if and only if f' is an immersion, and*
(11) *add more here.*

Proof. The properties $\mathcal{P}$ listed in the lemma are all stable under base change, hence if f' has property $\mathcal{P}$, then so does f. See Schemes, Lemmas 01JY and 02YC and Morphisms, Lemmas 01T4, 01TM, 02NK, 01V0, 02GA, 01W4, and 01WL.

The interesting direction in each case is therefore to assume that f has the property and deduce that f' has it too. By induction on the order of the thickening we may assume that $S \subset S'$ is a first order thickening, see discussion immediately following Definition 0.1.

Most of the proofs will use a reduction to the affine case. Let $U' \subset S'$ be an affine open and let $V' \subset X'$ be an affine open lying over U'. Let $U' = \mathrm{Spec}(A')$ and denote $I \subset A'$ be the ideal defining the closed subscheme $U' \cap S$. Say $V' = \mathrm{Spec}(B')$. Then $V' \cap X = \mathrm{Spec}(B'/IB')$. Setting $A = A'/I$ and $B = B'/IB'$ we get a commutative diagram

$$\begin{array}{ccccccccc} 0 & \longrightarrow & IB' & \longrightarrow & B' & \longrightarrow & B & \longrightarrow & 0 \\ & & \uparrow & & \uparrow & & \uparrow & & \\ 0 & \longrightarrow & IA' & \longrightarrow & A' & \longrightarrow & A & \longrightarrow & 0 \end{array}$$

with exact rows and $I^2 = 0$.

The translation of (1) into algebra: If $A \to B$ is surjective, then $A' \to B'$ is surjective. This follows from Nakayama's lemma (Algebra, Lemma 00DV).

The translation of (2) into algebra: If $A \to B$ is a finite type ring map, then $A' \to B'$ is a finite type ring map. This follows from Nakayama's lemma (Algebra, Lemma 00DV) applied to a map $A'[x_1, \ldots, x_n] \to B'$ such that $A[x_1, \ldots, x_n] \to B$ is surjective.

Proof of (3). Follows from (2) and that quasi-finiteness of a morphism which is locally of finite type can be checked on fibres, see Morphisms, Lemma 01TH.

Proof of (4). Follows from (2) and that the additional property of "being of relative dimension d" can be checked on fibres (by definition, see Morphisms, Definition 02NJ.

The translation of (5) into algebra: If $\Omega_{B/A} = 0$, then $\Omega_{B'/A'} = 0$. By Algebra, Lemma 00RV we have $0 = \Omega_{B/A} = \Omega_{B'/A'}/I\Omega_{B'/A'}$. Hence $\Omega_{B'/A'} = 0$ by Nakayama's lemma (Algebra, Lemma 00DV).

The translation of (6) into algebra: If $A \to B$ is unramified map, then $A' \to B'$ is unramified. Since $A \to B$ is of finite type we see that $A' \to B'$ is of finite type

by (2) above. Since $A \to B$ is unramified we have $\Omega_{B/A} = 0$. By part (5) we have $\Omega_{B'/A'} = 0$. Thus $A' \to B'$ is unramified.

Proof of (7). Follows by combining (2) with results of Lemma 09ZV and the fact that proper equals quasi-compact + separated + locally of finite type + universally closed.

Proof of (8). Follows by combining (2) with results of Lemma 09ZV and using the fact that finite equals integral + locally of finite type (Morphisms, Lemma 01WJ).

Proof of (9). As f is a monomorphism we have $X = X \times_S X$. We may apply the results proved so far to the morphism of thickenings $(X \subset X') \to (X \times_S X \subset X' \times_{S'} X')$. We conclude $X' \to X' \times_{S'} X'$ is a closed immersion by (1). In fact, it is a first order thickening as the ideal defining the closed immersion $X' \to X' \times_{S'} X'$ is contained in the pullback of the ideal $\mathcal{I} \subset \mathcal{O}_{S'}$ cutting out S in S'. Indeed, $X = X \times_S X = (X' \times_{S'} X') \times_{S'} S$ is contained in X'. Hence by Morphisms, Lemma 08S2 it suffices to show that $\Omega_{X'/S'} = 0$ which follows from (5) and the corresponding statement for X/S.

Proof of (10). If $f : X \to S$ is an immersion, then it factors as $X \to U \to S$ where $U \to S$ is an open immersion and $X \to U$ is a closed immersion. Let $U' \subset S'$ be the open subscheme whose underlying topological space is the same as U. Then $X' \to S'$ factors through U' and we conclude that $X' \to U'$ is a closed immersion by part (1). This finishes the proof. □